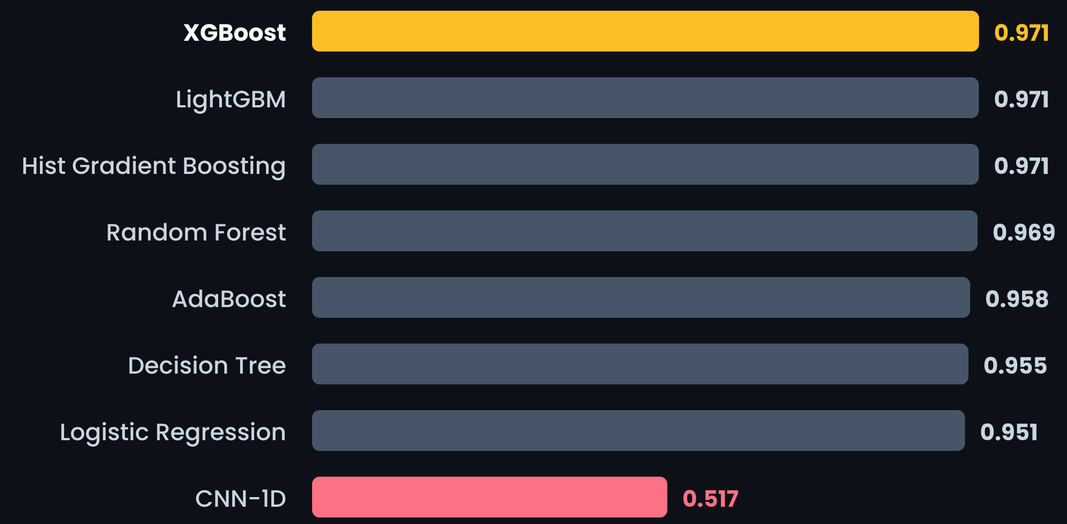
Model benchmark (AUC-ROC)

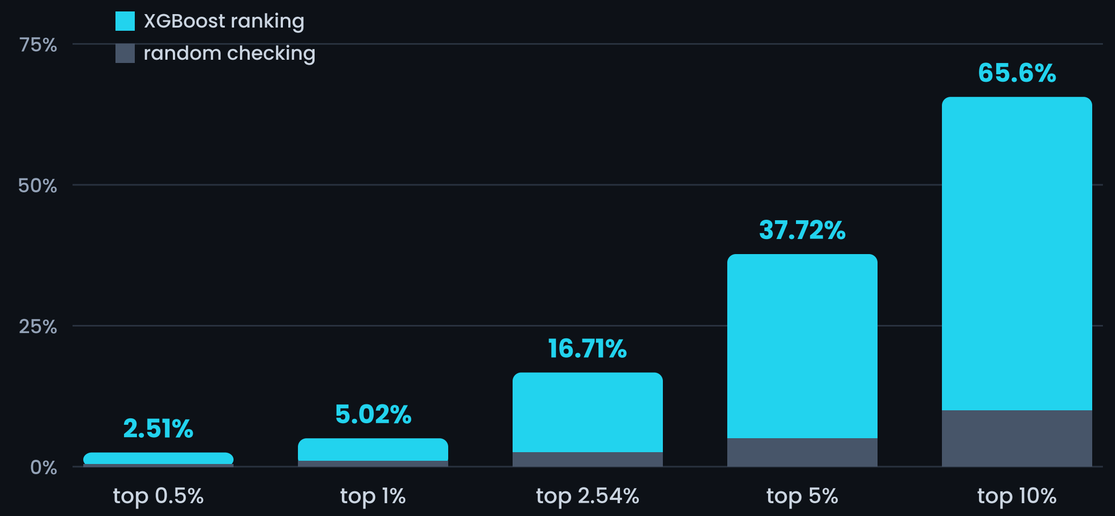
7 ML models on 1.02M held-out transactions · CNN on account sequences



XGBoost chosen for production · CNN-1D reported honestly as underperforming

Where the laundering sits

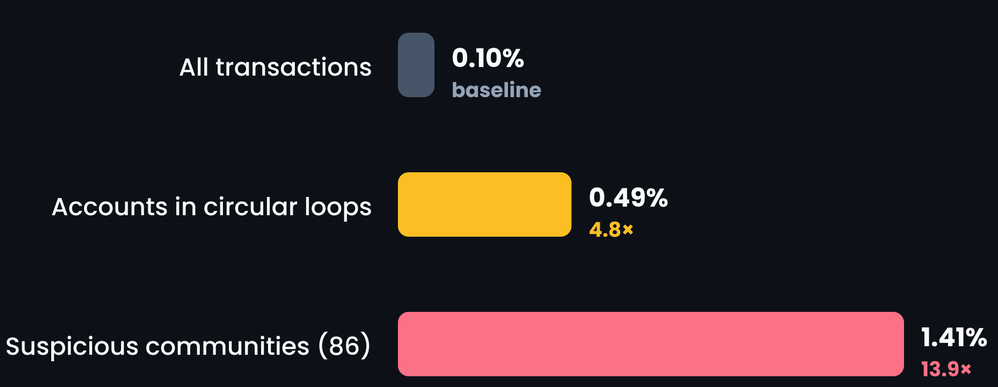
Share of all 5,177 fraud cases caught in the top-ranked transactions



Top 10% of transactions hold 65.6% of laundering: 6.6× better than random

Networks reveal hidden fraud

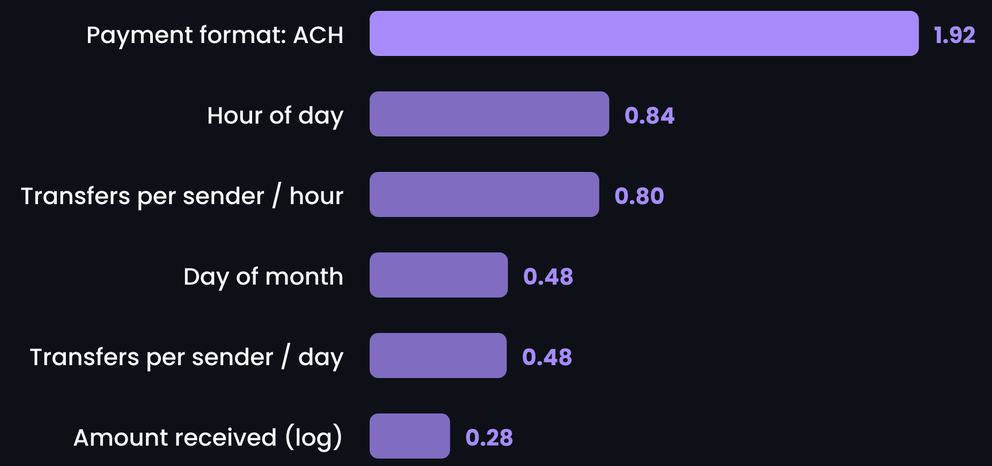
Fraud rate by where the account sits in the transaction graph



812 circular patterns (2 to 9 hops) found in a 5,000-account sample

What drives a fraud call (SHAP)

Mean |SHAP| impact of the top features in XGBoost



Timing and velocity features rank next to payment type: behaviour matters